FINAL PROJECT REPORT

DSC_672 DATA SCIENCE CAPSTONE PROJECT

BANK CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Name: Swetha Manupati, **Student id:** 2184430

1. Abstract:

Customer churn is a significant challenge in the banking industry, as losing customers directly impacts revenue and increases the cost of acquiring new ones. The objective of this project is to develop a machine learning model that can accurately predict whether a customer is likely to leave the bank.

This project follows a complete data science pipeline, including data exploration, preprocessing, exploratory data analysis (EDA), handling class imbalance using SMOTE, model training, and evaluation using classification metrics. The dataset consists of 10,000 bank customers with features such as credit score, age, balance, and customer activity.

Multiple machine learning models were implemented and compared, including Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine, and ensemble methods. After evaluating performance on both training and test data, a Hard Voting Ensemble model was selected as the final model.

The final model achieved approximately Recall = 0.715, Accuracy = 0.822, and F1-score = 0.620 on the test dataset, demonstrating a strong balance between identifying churners and maintaining overall accuracy. These results highlight the effectiveness of ensemble methods in improving model stability and generalization.

This model can support banks in proactively identifying at-risk customers and implementing targeted retention strategies, ultimately improving customer loyalty and reducing financial loss.

2. Introduction:

Customer churn refers to customers leaving a company's service, and in the banking industry, it is a critical problem. When customers leave, banks not only lose long-term revenue but also incur additional costs to acquire new customers. In fact, retaining existing customers is significantly more cost-effective than acquiring new ones, making churn prediction an important business priority.

I chose this project because churn prediction represents a real-world problem with direct financial impact. Banks manage large customer bases, and even small improvements in customer retention can lead to substantial savings. However, one of the main challenges is that it is difficult to identify which customers are at risk of leaving before the churn actually occurs. Without early detection, businesses are forced to react after the loss has already happened.

Machine learning provides an effective solution to this problem by analyzing customer data, identifying hidden patterns, and predicting churn in advance. This allows businesses to take proactive measures, such as targeted marketing or personalized offers, to retain high-risk customers.

The primary objective of this project is to develop a reliable and interpretable machine learning model that can accurately identify customers who are likely to churn. By doing so, the model can support data-driven decision-making and help banks improve customer retention strategies.

3. Literature Review:

Customer churn prediction has been widely studied using machine learning techniques, with a focus on improving predictive performance, handling class imbalance, and ensuring model interpretability. Prior research shows that ensemble models such as Random Forest and XGBoost outperform traditional methods by capturing complex patterns in customer behavior and improving accuracy and F1-score [1]. A major challenge identified is class imbalance, where churned customers are significantly fewer than non-churned ones. Techniques such as SMOTE have been successfully used to address this issue and improve the detection of churners. Key features influencing churn commonly include age, account balance, and number of products [1].

Another important area highlighted in the literature is model interpretability. While advanced models achieve higher accuracy, they often lack transparency. Studies comparing black-box models with simpler approaches like Logistic Regression, along with explainability techniques such as SHAP, show that interpretability is essential for building trust and enabling real-world adoption, especially in industries like banking [2].

Despite these advancements, gaps remain, including limited focus on cost-sensitive learning, scalability, and the translation of model performance into business impact. This project builds on prior work by combining ensemble methods, SMOTE, and

interpretability techniques, while emphasizing practical evaluation using recall and F1-score. The goal is not only to achieve strong predictive performance but also to deliver actionable insights for real-world decision-making.

4. Methodology:

4.1 Dataset Overview: Bank Customer Churn Dataset

The dataset consists of 10,000 bank customers with 12 features, including demographic, financial, and account-related attributes. The target variable is churn (0 = stayed, 1 = left).

The data is clean and complete, with no missing values and appropriate data types. Key statistics show:
Average age ≈ 39
Average credit score ≈ 650
Average balance ≈ 76,000
Most customers use 1–2 products

The target variable is imbalanced, with only ~20% churned customers, which requires special handling during modeling

4.2 Data Preparation

The dataset was prepared to ensure it is suitable for machine learning:

Feature Selection: Removed customer_id as it does not contribute to prediction
Encoding: Converted categorical variables (gender, country) into numerical format
Scaling: Applied feature scaling where required (e.g SVM)
Train-Test Split: Divided data into training and testing sets to evaluate generalization

These steps ensured the data was structured, consistent, and ready for modeling.

4.3 Exploratory Data Analysis (EDA)

EDA was performed to identify patterns and key drivers of churn.

Key Findings:

- Class Imbalance: ~20% churn rate
- Gender: Higher churn among female customers

- Country: Germany shows significantly higher churn
- Activity: Inactive customers are more likely to churn

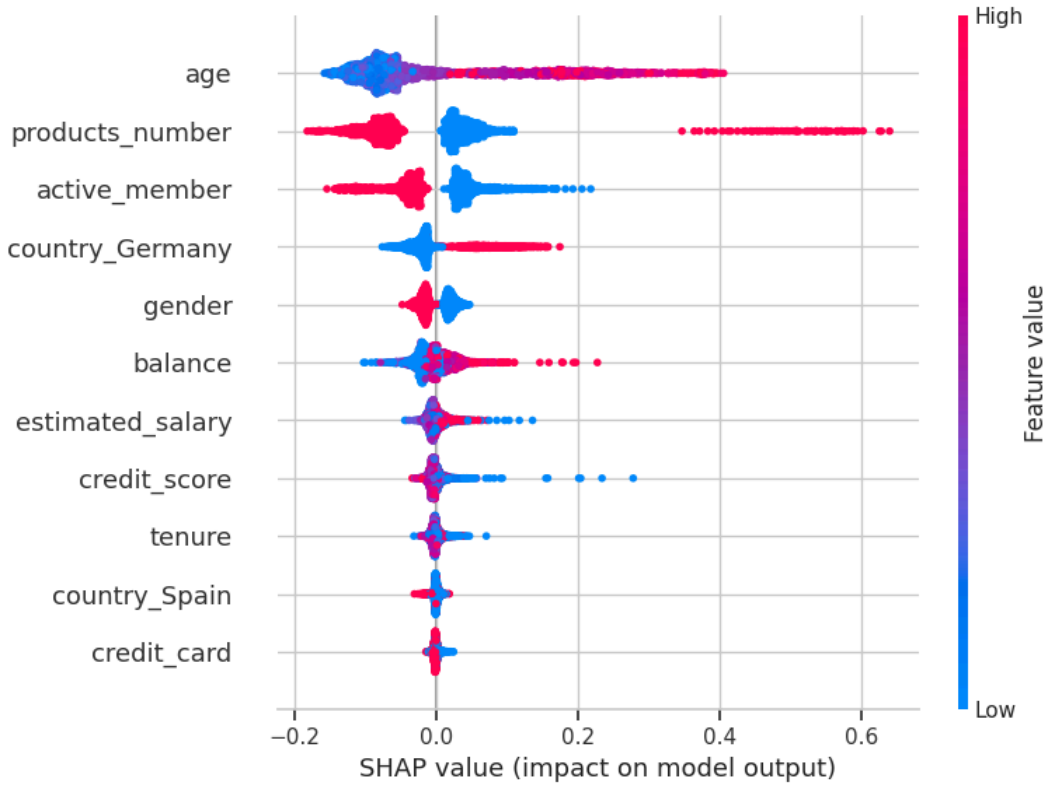
Numerical Insights:

- Customers with lower tenure and fewer products are more likely to leave
- Lower balances and engagement are associated with higher churn

These findings highlight that customer engagement and product usage are major factors influencing retention.

4.4 SHAP Analysis

In this project, interpretability was incorporated using SHAP analysis on the final models. The results confirmed that age, number of products, and customer activity status were the most influential features affecting churn predictions, with higher age and lower engagement increasing the likelihood of churn. These findings are consistent with prior research, which identified customer behavior and financial activity as key drivers. This demonstrates that combining high-performing models with interpretability techniques enables both accurate predictions and meaningful business insights, making the model more practical for real-world banking applications.



4.5 Handling Class Imbalance

Since the dataset was imbalanced, SMOTE (Synthetic Minority Oversampling Technique) was applied to the training data to improve minority class representation. Before applying SMOTE, the class distribution was highly skewed, with **6,370 non-churned customers and 1,630 churned customers**. After applying SMOTE, the dataset was balanced with **6,370 instances in each class**.

SMOTE works by generating synthetic samples for the minority class (churned customers), rather than duplicating existing data. This balancing allowed the models to better learn patterns associated with churn, significantly improving recall. As a result, the model became more effective at identifying at-risk customers, which is critical for real-world business applications.

5. Models Trained:

To identify the most effective approach for predicting customer churn, a diverse set of machine learning models was implemented and compared. These included Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naive Bayes, and a Neural Network (MLP). Logistic Regression was used as a baseline model due to its simplicity and interpretability, while Decision Trees and KNN provided intuitive, non-linear approaches. Ensemble methods such as Random Forest, Gradient Boosting, and XGBoost were included for their ability to improve predictive performance and reduce variance.

Each model has its strengths and limitations. Simpler models like Logistic Regression and Naive Bayes are fast and easy to interpret but may fail to capture complex relationships. More advanced models such as Gradient Boosting, XGBoost, and Neural Networks are capable of modeling complex patterns and achieving higher accuracy but require careful tuning to avoid overfitting. SVM performs well in high-dimensional spaces, while Random Forest improves stability compared to single decision trees. This variety of models ensured a comprehensive comparison, enabling the selection of the most suitable model for churn prediction based on both performance and generalization.

6. Hyperparameter Tuning and Overfitting Analysis

After training the initial set of nine machine learning models, significant overfitting was observed, particularly in tree-based models such as Decision Tree and Random Forest. These models achieved very high performance on the training data but showed reduced performance on the test data, indicating poor generalization. In addition, due to class imbalance, most models achieved high overall accuracy (~80–85%) but failed to correctly identify churned customers, resulting in low recall for the minority class.

This highlighted the limitations of baseline models and the need for further optimization.

To address these issues, hyperparameter tuning was performed using GridSearchCV, focusing on reducing model complexity and improving recall. Cross-validation was used during this process to evaluate model performance across multiple data splits, ensuring more reliable results and reducing overfitting. Parameters such as maximum tree depth, number of estimators, learning rate, and regularization strength were optimized across different models. This process helped reduce overfitting and improved model stability on unseen data. However, the improvement in detecting churners remained limited, indicating that hyperparameter tuning alone was not sufficient. This led to the implementation of additional strategies, including threshold tuning, class weighting, and ensemble methods, to further enhance model performance.

7. Model Improvement and Optimization

To address overfitting observed in baseline models, improved versions were developed using **regularization and controlled model complexity**. For tree-based models such as Gradient Boosting and XGBoost, parameters including maximum depth, minimum samples per leaf, and learning rate were reduced to prevent overly complex trees. Similarly, the neural network architecture was simplified and regularized, while models like Logistic Regression and SVM inherently benefited from regularization. These improvements reduced the gap between training and test performance, resulting in better generalization and more stable predictions.

Beyond model simplification, three key strategies were applied to improve churn detection: **threshold optimization, class weighting, and ensemble learning**. These strategies specifically focused on improving the detection of churners (minority class) while maintaining balanced overall performance.

To further evaluate model behavior and confirm the presence of overfitting, training performance was analyzed across all models. The results show that simpler models such as Logistic Regression and Naive Bayes achieved moderate and consistent performance, while more complex models like Decision Tree and Random Forest achieved perfect training accuracy (100%), indicating clear overfitting. In contrast, models such as Gradient Boosting, XGBoost, and Support Vector Machine maintained strong performance without overfitting, demonstrating better generalization. These findings highlight the importance of model regularization and justify the use of improved models and ensemble techniques for more reliable predictions.

Training Results Table:

Model	Precision	Recall	F1-Score	Accuracy
Logistic Regression	0.72	0.70	0.71	0.71
Decision Tree	1.00	1.00	1.00	1.00
Random Forest	1.00	1.00	1.00	1.00
Gradient Boosting (Improved)	0.85	0.83	0.84	0.84
XGBoost (Improved)	0.84	0.81	0.83	0.83
Support Vector Machine	0.83	0.85	0.84	0.84
K Nearest Neighbors	0.85	0.98	0.91	0.90
Naive Bayes	0.72	0.70	0.72	0.72
Neural Network (Improved)	0.92	0.98	0.95	0.95

To further validate model performance, test results were analyzed to assess generalization on unseen data. The results show that models such as Decision Tree and Random Forest, despite achieving perfect training performance, experienced a noticeable drop in test performance, confirming overfitting. Simpler models like Logistic Regression and Naive Bayes maintained consistent but relatively lower performance. In contrast, models such as Gradient Boosting, XGBoost, and Support Vector Machine demonstrated more stable and balanced results, with strong recall and good overall accuracy. These models were better able to generalize to new data, making them more suitable for real-world deployment.

Overall, the test results reinforce the importance of controlling model complexity and selecting models that balance performance and generalization. However, model performance can still be improved, which motivated the use of additional strategies to achieve more robust and reliable predictions.

Test Results Table:

Model	Precision	Recall	F1-Score	Accuracy
Logistic Regression	0.39	0.70	0.50	0.72
Decision Tree	0.43	0.51	0.47	0.77
Random Forest	0.61	0.61	0.61	0.84
Gradient Boosting (Improved)	0.54	0.70	0.61	0.82
XGBoost (Improved)	0.53	0.72	0.61	0.81
Support Vector Machine	0.49	0.72	0.59	0.79
K Nearest Neighbors	0.39	0.65	0.49	0.72
Naive Bayes	0.39	0.68	0.49	0.71
Neural Network (Improved)	0.46	0.61	0.52	0.77

7.1 Strategy 1: Decision Threshold Optimization

Instead of using the default threshold (0.5), multiple thresholds (0.3-0.6) were evaluated using predicted probabilities from the Gradient Boosting model.

Threshold	Recall	Accuracy
0.3	0.880	0.665
0.4	0.786	0.767
0.5	0.703	0.818
0.6	0.634	0.851

Lower thresholds significantly improved recall by identifying more churners, while higher thresholds improved precision and accuracy. This demonstrates a clear trade-off, allowing threshold selection based on business priorities.

7.2 Strategy 2: Class Weight Adjustment

Class weights were applied to give more importance to churners during training.

Model	Recall	Accuracy
Logistic Regression (class weight 1:3)	0.931	0.439
Logistic Regression(Balanced)	0.698	0.716
Gradient Boosting(class weight)	0.703	0.818

While higher weights improved recall, they significantly reduced precision and accuracy. This shows that excessive weighting leads to many false positives. Moderate weighting improved performance, but alone was not sufficient.

7.3 Strategy 3: Voting Ensemble (Final Approach)

A Voting Ensemble combining Gradient Boosting, Random Forest, and Logistic Regression was implemented.

Voting Method	Recall	F1- Score	Accuracy
Hard Voting	0.715	0.620	0.822
Soft Voting	0.715	0.605	0.810

The Hard Voting model provided the best balance across all metrics by combining the strengths of multiple models, reducing variance, and improving generalization. Each strategy targeted a key limitation: threshold tuning improved recall, class weighting addressed imbalance (with trade-offs), and ensemble learning provided the best overall balance.

The Hard Voting Ensemble was selected as the final model, achieving strong recall while maintaining good accuracy and F1-score.

8. Results and Discussion

8.1 Model Performance Overview

The performance of all models and strategies was evaluated using recall, precision, F1-score, and accuracy. The results highlight a clear trade-off between detecting churners (recall) and maintaining overall prediction accuracy.

From the final comparison, class-weighted Logistic Regression (1:3) achieved the highest recall (0.931), indicating its strong ability to detect churners. However, this came at a significant cost to precision (0.257) and accuracy (0.439), resulting in many false positives. Similarly, threshold optimization at 0.3–0.4 improved recall (up to 0.880) but reduced overall accuracy, making it less suitable for practical deployment.

Among baseline models, Support Vector Machine (SVM) achieved the highest recall (0.722) but had lower accuracy (0.793) and F1-score (0.587), indicating weaker balance across metrics. In contrast, Gradient Boosting and XGBoost provided more stable and balanced performance, with recall around 0.70 and accuracy above 0.81.

```
--- RANKING BY RECALL (Best for Catching Churners) ---
```

	Strategy	Model	Recall	Precision	F1-Score	Accuracy
8	Class Weights	Logistic Regression (Class Weight 1:3)	0.931204	0.257298	0.403191	0.4390
1	Threshold Optimization	GB Improved	0.879607	0.366053	0.516968	0.6655
2	Threshold Optimization	GB Improved	0.786241	0.457797	0.578662	0.7670
13	Voting Ensemble	SOFT VOTING	0.714988	0.524324	0.604990	0.8100
12	Voting Ensemble	HARD VOTING	0.714988	0.546992	0.619808	0.8215
0	Baseline (GB Improved)	Gradient Boosting (Improved)	0.702703	0.540643	0.611111	0.8180
9	Voting Ensemble	GB	0.702703	0.540643	0.611111	0.8180
5	Class Weights	Gradient Boosting (Class Weight 1:1)	0.702703	0.540643	0.611111	0.8180
3	Threshold Optimization	GB Improved	0.702703	0.540643	0.611111	0.8180
6	Class Weights	Gradient Boosting (Class Weight 1:2)	0.702703	0.540643	0.611111	0.8180
10	Voting Ensemble	RF	0.700246	0.557730	0.620915	0.8260
7	Class Weights	Logistic Regression (Balanced)	0.697789	0.389041	0.499560	0.7155
11	Voting Ensemble	LR	0.697789	0.389041	0.499560	0.7155
4	Threshold Optimization	GB Improved	0.633907	0.632353	0.633129	0.8505

8.2 Strategy Comparison and Key Insights

The three improvement strategies addressed different aspects of the problem:

- Threshold Optimization significantly improved recall but introduced more false positives, reducing precision and accuracy.
- Class Weighting helped address class imbalance but showed diminishing returns when weights were too aggressive.

- Voting Ensemble provided the most consistent and balanced performance across all evaluation metrics.

The **Hard Voting Ensemble**, combining Gradient Boosting, Random Forest, and Logistic Regression, achieved:

- **Recall:** 0.715
- **Precision:** 0.547
- **F1-score:** 0.620
- **Accuracy:** 0.822

This performance demonstrates a strong balance between identifying churners and maintaining reliable overall predictions.

8.3 Final Model Selection and Justification

Although some models achieved higher recall or accuracy individually, they lacked balance across all metrics. For example, threshold tuning at 0.6 achieved the highest accuracy (**0.851**) but significantly reduced recall (**0.634**), while SVM achieved higher recall but lower overall performance.

The **Hard Voting Ensemble** was selected as the final model because it offers:

- Balanced performance across recall, precision, and accuracy
- Improved generalization compared to individual models
- Reduced overfitting through model combination
- Greater robustness for real-world deployment

8.4 Business Interpretation

From a business perspective, recall is critical because it reflects the model's ability to identify customers who are likely to churn. The selected model successfully captures approximately **71.5% of churners**, enabling targeted retention strategies.

At the same time, maintaining good precision and accuracy ensures that marketing resources are not wasted on too many false positives. Therefore, the Voting Ensemble provides the best practical balance between **maximizing customer retention and minimizing unnecessary costs**, making it suitable for deployment in real-world banking systems.

8. Conclusion

This project successfully developed a machine learning system to predict customer churn using a structured and end-to-end data science approach. Multiple models were trained and evaluated, and key challenges such as class imbalance and overfitting were effectively addressed using SMOTE, regularization, and model improvement strategies.

The results demonstrated that model performance varies depending on the chosen strategy. Recall ranged from approximately **0.63 to 0.93**, highlighting the trade-off between detecting churners and maintaining overall accuracy. While some approaches achieved very high recall, they suffered from low precision and accuracy, making them less practical for real-world use.

The **Hard Voting Ensemble** was selected as the final model because it provides the best balance across all evaluation metrics (**Recall: 0.715, Accuracy: 0.822, F1-score: 0.620**). This balance is critical in business applications, where both identifying churners and minimizing false positives are important. Overall, the model is robust, generalizes well, and is suitable for real-world deployment.

9. Future Work:

Although the current model performs well, several improvements can further enhance its effectiveness. One key area is improving precision, as reducing false positives will make retention strategies more cost-efficient. This can be achieved through advanced techniques such as cost-sensitive learning, better feature engineering, or optimizing decision thresholds based on business objectives.

Another important improvement is the use of larger and more diverse datasets. Collecting more churn-related data will help reduce class imbalance and allow the model to learn more representative patterns. Additionally, incorporating behavioral features such as transaction history or customer interactions could significantly improve predictive performance.

In real-world applications, models must also be continuously updated. Regular retraining is essential to handle data drift and changing customer behavior over time. Future work could also explore more advanced models such as LightGBM, CatBoost, or stacking ensembles, as well as deploying the model into a production environment using APIs or dashboards for real-time predictions.

Overall, while the current model provides strong and reliable performance, continuous improvement, monitoring, and integration with business processes will further enhance its value and impact.

10. References:

1. Bank customer churn prediction using machine learning. Retrieved from https://www.researchgate.net/publication/387807835_Bank_Customer_Churn_Prediction_Using_Machine_Learning
2. Explainable machine learning models for bank customer churn prediction. Retrieved https://www.researchgate.net/publication/371139221_Explainable_Machine_Learning_Models_for_Bank_Customer_Churn_Prediction